fleet-equipment-data-analysis

📌 Project Overview

This project is part of my IBM Data Analytics coursework, where I analysed a fleet equipment inventory dataset using Microsoft Excel Pivot Tables.

The goal was to transform raw data into meaningful insights, identify patterns in equipment distribution, and derive business recommendations for better resource management.

📊 Objectives Clean and organize raw fleet data Analyse equipment distribution across departments Identify trends using pivot tables Generate actionable business insights 🛠️ Tools & Technologies Used Microsoft Excel Pivot Tables Data Cleaning Techniques Data Aggregation & Analysis

📈 Key Insights

1. Department-wise Equipment Distribution the Transportation department holds the majority of equipment (~1221 units), significantly higher than other departments. Departments like Permitting Services and Sheriff's Office have comparatively fewer assets.

👉 Insight: Indicates a strong operational focus on transportation activities.

2. Equipment Type Analysis Heavy-duty and transportation-related equipment dominate the dataset. Categories like Compact Utility Vehicles (CUVs) are less common.

👉 Insight: Suggests reliance on high-capacity equipment for operational needs.

3. Uneven Resource Allocation Some departments have very limited equipment compared to others.

👉 Insight:

4. Could indicate lower demand Or potential inefficiency in resource allocation 4. Multi-Department Equipment Usage Certain equipment types are used across multiple departments.

👉 Insight: These assets are multi-functional and can be optimized for shared usage.

5. Operational Efficiency Opportunities High concentration of equipment in one department may lead to: Better coordination OR potential underutilization

👉 Insight: Monitoring usage patterns can improve cost efficiency.

💡 Business Recommendations Optimize fleet distribution across departments Track equipment utilization rates Identify underused assets for cost reduction Implement shared resource strategies Build dashboards for real-time monitoring (Power BI / Tableau)

▯ Skills Demonstrated Data Cleaning & Preparation Pivot Table Analysis Data Interpretation Business Insight Generation Data Storytelling

🚀 Project Summary

Analysed fleet equipment inventory using Excel Pivot Tables to uncover distribution patterns, resource allocation gaps, and opportunities for operational optimization.

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